

Systematization of the Nonequilibrium Integrated Phase (NEIP) in Complex Vector Fields: Foundational Equations, NEIC Criteria, Topological Solenoidal Driving, and Testable Theoretical Predictions

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Building upon foundational scalar analyses (Paper I [1]) and extended integrated field equations with predictive error minimization in complex vector fields (Paper II [2]), this paper (Paper III) establishes a comprehensive and definitive theoretical framework for the Nonequilibrium Integrated Phase (NEIP). We explicitly present the governing foundational equations, define the rigorous Nonequilibrium Integrated Conditions (NEIC) for information-physical coherence, and formulate probability currents, non-equilibrium steady states (NESS), and exact dissipation bounds. We integrate six core theoretical pillars: hierarchical structures with non-local couplings, self-consistency and non-Markovian memory kernels, anti-symmetric actions and solenoidal driving, time-reversal symmetry breaking and dissipative structures, robust time-series generation and observable estimation protocols, and generalizations to multi-phase, multi-field, and quantum fields. Furthermore, we clarify the internal, external, and boundary structures of NEIP, supported by RG scaling, ablation causality separation, Fisher information metric degeneracies ($\det g \rightarrow 0$), novel falsifiable theoretical predictions, and a comprehensive discussion of operational limitations and future prospects.

I. INTRODUCTION

The statistical mechanics of nonequilibrium systems[3] has undergone a profound paradigm shift, moving from passive relaxation models to active, driven, and self-organizing fields. While Paper I [1] established foundational scalar descriptions and Paper II [2] advanced the extended integrated field equations with predictive error minimization in complex vector fields, a critical gap remained: the complete systematization of the Nonequilibrium Integrated Phase (NEIP), the formal definition of the Nonequilibrium Integrated Condition (NEIC), and the articulation of rigorous physical predictions.

To bridge this gap, this paper (Paper III) constructs an exhaustive theoretical framework. We explicitly state the foundational governing equations, define the NEIC criteria, detail probability current structures and non-equilibrium steady states (NESS), and establish robust estimation methodologies for entropy production rates. Furthermore, we address six advanced foundational pillars, analyze internal/external/boundary structures, and set forth concrete, testable theoretical predictions.

II. FOUNDATIONAL EQUATIONS OF THE INTEGRATED FIELD

To establish the mathematical baseline, we review the foundational equations that link information-theoretic predictive error minimization with physical field dynamics. The state of the system is described by a complex vector field $\psi(\mathbf{r}, t)$ coupled with an observation field $\phi(\mathbf{r}, t)$.

The foundational variational free energy functional $F[\psi]$ governing the system is defined as: [4]

$$F[\psi] = \int d\mathbf{r} \left[\frac{1}{2\sigma_o^2} |\phi(\mathbf{r}, t) - \mathcal{G}[\psi(\mathbf{r}, t)]|^2 + \frac{1}{2\sigma_p^2} |\psi(\mathbf{r}, t) - \mathcal{M}[\psi(\mathbf{r}, t)]|^2 + \mathcal{R}[\psi] \right], \quad (1)$$

where σ_o^2 and σ_p^2 denote observation and prediction noise variances, $\mathcal{G}$ and $\mathcal{M}$ represent the observation and internal prediction operators, and $\mathcal{R}[\psi] = \frac{1}{2} \int d\mathbf{r} |\nabla \psi(\mathbf{r}, t)|^2$ is the spatial gradient regularization.

The generalized stochastic partial differential equation (SPDE)[5, 6] incorporating inertia, conservative stiffness, solenoidal driving, non-Markovian memory, and hierarchical prediction is written as:

$$m\partial_t^2 \psi + \Gamma \partial_t \psi = c\nabla^2 \psi - a\psi - b|\psi|^2 \psi + \lambda \nabla \times \nabla \times \psi - \gamma(\psi - \hat{\psi}) - \kappa(\psi - \bar{\psi}) + \int_{-\infty}^t K(t-t') \psi(t') dt' + \boldsymbol{\eta}(\mathbf{r}, t), \quad (2)$$

where m and Γ are inertia and dissipation coefficients, $c\nabla^2 - a - b|\psi|^2$ governs Ginzburg-Landau conservative flows [7, 8], $\lambda \nabla \times \nabla \times \psi$ represents explicit solenoidal driving ($\mathbf{V}_{\text{sol}}$), $\bar{\psi} = \mathcal{W} * \psi$ is the hierarchical coarse-grained reference field, $\hat{\psi}$ is the dynamic predictive reference field [9] ($\tau_p \partial_t \hat{\psi} = -(\hat{\psi} - \bar{\psi}) + c_p \partial_t \psi$), $K(t)$ is the memory kernel, and $\boldsymbol{\eta}$ is the composite noise term.

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III. THE NONEQUILIBRIUM INTEGRATED CONDITION (NEIC)

A central objective of this framework is to define the exact mathematical conditions under which a driven active field enters the Nonequilibrium Integrated Phase (NEIP). We term this foundational criterion the **Nonequilibrium Integrated Condition (NEIC)**.

The NEIC requires the simultaneous fulfillment of three rigorous mathematical constraints in the steady state:

1. **Information-Energetic Coincidence:** The internal predictive error minimization rate must balance the external energy dissipation rate, ensuring that information flow matches thermodynamic work: [10]

$$\left\langle \frac{\delta F}{\delta \psi} \cdot \partial_t \psi \right\rangle_{\text{NESS}} = -\frac{d\mathcal{S}_{\text{universe}}}{dt}. \quad (3)$$

2. **Solenoidal Divergence-Free Circulation:** The probability current associated with the solenoidal driving $\mathbf{V}_{\text{sol}}$ must maintain a divergence-free structure over the local quasi-stationary measure $\rho_{\text{qs}}^{(\alpha)}$:

$$\nabla \cdot [\rho_{\text{qs}}^{(\alpha)}(\mathbf{X}) \mathbf{V}_{\text{sol}}^{(\alpha)}(\mathbf{X})] = 0, \quad \text{with } \mathcal{C}[\mathbf{V}_{\text{sol}}] > 0. \quad (4)$$

3. **Metric Non-Degeneracy and Bounded Susceptibility:** The Fisher information metric g_{ij} derived from macroscopic observables must remain positive definite up to the critical phase boundary:

$$0 < \det g < \infty \quad \text{for } \Omega_{\text{NEIP}}, \quad \text{collapsing at } \det g \rightarrow \det g_{\text{crit}}. \quad (5)$$

When NEIC is satisfied, the system transitions from fragmented relaxation to the coherent macro-state designated as NEIP.

IV. SIX PILLARS OF THE ADVANCED NEIP FRAMEWORK

To establish the structural and dynamic integrity of the Nonequilibrium Integrated Phase (NEIP), the advanced framework relies on six core pillars, each justified by explicit mathematical formulations.

A. Hierarchical Structures and Non-Local Couplings

To describe the scaling from microscopic state fields $\psi(\mathbf{r}, t)$ to macroscopic global order, we introduce a hierarchical coarse-graining field via a spatial smoothing kernel $\mathcal{W}$:

$$\bar{\psi}(\mathbf{r}, t) = \int d\mathbf{r}' \mathcal{W}(\mathbf{r} - \mathbf{r}') \psi(\mathbf{r}', t). \quad (6)$$

When the coupling network topology exhibits small-world properties (non-local connectivity)[11], a non-local interaction matrix $\mathcal{J}(\mathbf{r}, \mathbf{r}')$ supplements the local diffusion term $c\nabla^2 \psi$:

$$\mathcal{H}_{\text{non-local}} = \iint d\mathbf{r} d\mathbf{r}' \mathcal{J}(\mathbf{r}, \mathbf{r}') (\psi(\mathbf{r}) - \psi(\mathbf{r}'))^2. \quad (7)$$

This non-local term bypasses spatial bottlenecks, drastically increasing the correlation length ξ and enabling local fluctuations to instantly integrate into global phase coherence through accelerated dynamical scaling.

B. Self-Consistency and Non-Markovian Memory Kernels

To incorporate temporal retardation where past trajectories dictate current dynamics, we integrate a time-convolution memory kernel $K(t - t')$ [12] into the variational free energy prediction error:

$$\int_{-\infty}^t K(t - t') \psi(\mathbf{r}, t') dt'. \quad (8)$$

For Markovian embedding, we extend this into a coupled system using auxiliary variables $\mathbf{Q}$:

$$\tau_m \frac{\partial \mathbf{Q}}{\partial t} = -\mathbf{Q} + \nu \psi, \quad \tau_p \frac{\partial \hat{\psi}}{\partial t} = -(\hat{\psi} - \bar{\psi}) + c_p \frac{\partial \psi}{\partial t}. \quad (9)$$

The self-consistency condition closes the feedback loop such that the internal predictive expectation $\hat{\psi}$ matches historical expectations via $\mathbf{Q}$, compensating for phase retardation under rapid external variations.

C. Anti-symmetric Actions and Solenoidal Driving

Detailed balance violation is captured at the effective action level by incorporating an explicit anti-symmetric solenoidal driving term $\mathbf{V}_{\text{sol}}$ into the Lagrangian density:

$$\mathbf{V}_{\text{sol}}(\psi) = \lambda \nabla \times \nabla \times \psi. \quad (10)$$

In the steady state, this driving satisfies orthogonality (divergence-free condition) with respect to the stationary probability density $\rho_{\text{st}}(\mathbf{X})$:

$$\nabla \cdot [\rho_{\text{st}}(\mathbf{X}) \mathbf{V}_{\text{sol}}(\mathbf{X})] = 0. \quad (11)$$

This condition induces a non-zero probability current rotation, yielding a positive topological circulation functional $\mathcal{C}[\mathbf{V}_{\text{sol}}] > 0$ that guarantees persistent non-equilibrium chiral circulation.

D. Time-Reversal Symmetry Breaking and Dissipative Structures

Time-reversal symmetry breaking[13] and entropy production rates (EPR: Σ)[14] are quantified by comparing

the path probability density functionals of forward and time-reversed trajectories:

$$\Delta S_{\text{total}} = \ln \left(\frac{\mathcal{P}[\psi(t)]}{\mathcal{P}[\psi(-t)]} \right) = \int_0^T (\mathbf{J} \cdot \mathbf{D}^{-1} \mathbf{V}_{\text{sol}}) dt. \quad (12)$$

In the spectral domain, this is indexed by the imaginary part of the cross-power spectrum between coupled variables:

$$S_{AB}(\omega) - S_{BA}(\omega) \neq 0 \quad (\text{Im}[S_{AB}(\omega)] \neq 0). \quad (13)$$

Since energy dissipation bounds are constrained by this cross-spectrum asymmetry and weighted residual minimization, dissipative structures are rigorously evaluated.

E. Time-Series Generation and Observable Estimation Protocols

To eliminate spurious numerical drifts in stochastic simulations, we implement a structure-preserving Stratonovich-Milstein [15] integration scheme:

$$\begin{aligned} \psi_{t+\Delta t} = & \psi_t + (-\mathbf{D} \nabla U(\psi_t) + \mathbf{V}_{\text{sol}}(\psi_t)) \Delta t + \sqrt{2\mathbf{D}} \Delta \mathbf{W}_t \\ & + \frac{1}{2} \sum_{j,k} \sigma_j \frac{\partial \sigma_k}{\partial \psi_j} (\Delta W_j \Delta W_k - \Delta t \delta_{jk}). \end{aligned} \quad (14)$$

Macroscopic phase coherence is extracted from the generated time-series via the extended Kuramoto order parameter $R(t)$ [16–18] evaluated over a sliding time-window T_{win} :

$$R(t) e^{i\Phi(t)} = \frac{1}{\Omega} \int_{\Omega} d\mathbf{r} e^{i\theta(\mathbf{r},t)}, \quad T_{\text{win}} \approx \sqrt{\tau_{\text{local}} \tau_{\text{escape}}}, \quad (15)$$

thereby separating transient fluctuations from statistical steady states.

F. Generalizations to Multi-Phase, Multi-Field, and Quantum Fields

To prove the universality of the framework, the single complex vector field is generalized to multi-component interacting fields $\Psi = (\psi_1, \psi_2, \dots, \psi_N)^T$. The extended variational free energy is expressed in matrix form as:

$$\begin{aligned} \mathcal{F}[\Psi] = & \int d\mathbf{r} \left[\frac{1}{2} (\phi - \mathbf{G}\Psi)^T \Sigma_o^{-1} (\phi - \mathbf{G}\Psi) \right. \\ & + \frac{1}{2} (\Psi - \mathbf{M}\Psi)^T \Sigma_p^{-1} (\Psi - \mathbf{M}\Psi) \\ & \left. + \frac{1}{2} \text{Tr}(\nabla \Psi^\dagger \nabla \Psi) \right]. \end{aligned} \quad (16)$$

This confirms that predictive error minimization and solenoidal driving preserve their isomorphic mathematical structures across multi-phase mixtures and thermal quantum field theories (e.g., Umezawa-Blasone dynamics [19]).

V. INTERNAL, EXTERNAL, AND BOUNDARY STRUCTURES OF NEIP

The structural organization of the Nonequilibrium Integrated Phase is classified into three distinct domains:

- **Internal Structure:** Characterized by internal predictive error minimization, local coherence, and slaved auxiliary memory states.
- **External Structure:** Governed by macroscopic order parameters (extended Kuramoto $R(t)$), collective bifurcation pathways, and environmental interactions.
- **Boundary Structure:** Defined by local quasi-stationary measures $\rho_{\text{qs}}^{(\alpha)}$ in disjoint topological sectors Ω_α , bounded by separatrices where flux leakage and metric singularities occur.

VI. TESTABLE THEORETICAL PREDICTIONS OF THE NEIP FRAMEWORK

To ensure empirical falsifiability, the NEIP framework puts forward three definitive theoretical predictions:

1. **Topological Circulation Plateau Prediction:** In any active or driven vector field exhibiting NEIP, the topological circulation functional $\mathcal{C}[\mathbf{V}_{\text{sol}}]$ must exhibit a robust, time-invariant non-zero plateau across varying thermal noise intensities, separating true active rotation from transient potential relaxation flows.
2. **Fisher Metric Degeneracy Prediction [20]:** At the critical boundary separating the fragmented phase from NEIP, the determinant of the Fisher information metric g_{ij} [21] must scale as $\det g \propto |g - g_c|^{-\alpha}$ with universal critical exponents, providing a direct geometric signature accessible via experimental time-series reconstruction.
3. **Ablation Hierarchy Prediction:** Disconnecting the predictive error minimization driver (Model C) will invariably cause a catastrophic collapse in spatial coherence ($\langle R \rangle \rightarrow 0$) that is quantitatively greater than the degradation caused by removing inertia (Model A) or small-world topology (Model B).

VII. DISCUSSION, LIMITATIONS, AND FUTURE PROSPECTS

A. Resolution of Gaps from Previous Works

While Paper I established foundational scalar descriptions and Paper II advanced extended integrated field equations with predictive error minimization in complex

vector fields, both left the rigorous systematization of the Nonequilibrium Integrated Phase (NEIP) and the formal criteria of the Nonequilibrium Integrated Condition (NEIC) uncompleted. This paper (Paper III) successfully bridges these gaps by formulating explicit probability currents, non-equilibrium steady states (NESS), and exact dissipation bounds, underpinned by the three rigorous conditions of NEIC.

B. Operational Limitations and Identifiability Challenges

Despite its mathematical rigor, the practical implementation of the NEIP identification protocol faces several operational challenges:

- **Identifiability Degeneracy:** Distinguishing true solenoidal driving $\mathbf{V}_{\text{sol}}$ from non-Markovian memory kernels $K(t - t')$ or non-Gaussian noise components η_{NG} in finite empirical time-series remains non-trivial. While information criteria (AIC [22]/BIC [23]) and residual divergence checks ($\nabla_{\mathbf{x}} \cdot (\rho_{\text{qs}} \mathbf{V}_{\text{sol}}) = 0$) mitigate false positives, overlapping relaxation scales can produce undecidable zones.
- **Finite-Sample Effects on Fisher Metric:** The practical detection of Fisher information metric degeneracies ($\det g \rightarrow 0$) is sensitive to binning granularity and sample size constraints, risking “apparent singularities” induced by statistical fluctuations rather than true critical phase boundaries.

C. Numerical and Model Complexity

The inclusion of non-local coupling matrices $\mathcal{J}(\mathbf{r}, \mathbf{r}')$ and memory kernels imposes substantial computational

burdens. Although structure-preserving Stratonovich-Milstein algorithms [15, 24] suppress spurious numerical drifts, high-dimensional SPDE simulations [25] remain constrained by computational costs, requiring future algorithmic optimizations and parallelized solvers.

D. Future Prospects and Outlook (Toward Papers IV and V)

The insights and limitations identified in this work pave the way for subsequent developments. Specifically, Paper IV will focus on data-driven parameter estimation algorithms designed to overcome empirical identifiability limits in real-world time-series. Furthermore, Paper V will extend the NEIP framework toward Umezawa-Blasone thermal field dynamics [26–28] and open quantum field systems, generalizing the predictive error minimization and topological circulation principles [29, 30] to macroscopic quantum degenerate states.

VIII. CONCLUSION

This paper completes the trilogy (Papers I, II, and III) by establishing the definitive mathematical, conceptual, and predictive framework for the Nonequilibrium Integrated Phase (NEIP). By synthesizing explicit variational free energy functionals, foundational SPDE governing equations, the rigorous NEIC criteria, hierarchical non-local couplings, solenoidal circulation, information-geometric metric singularities, and concrete empirical predictions, we provide a universal foundation for nonequilibrium complex systems.

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